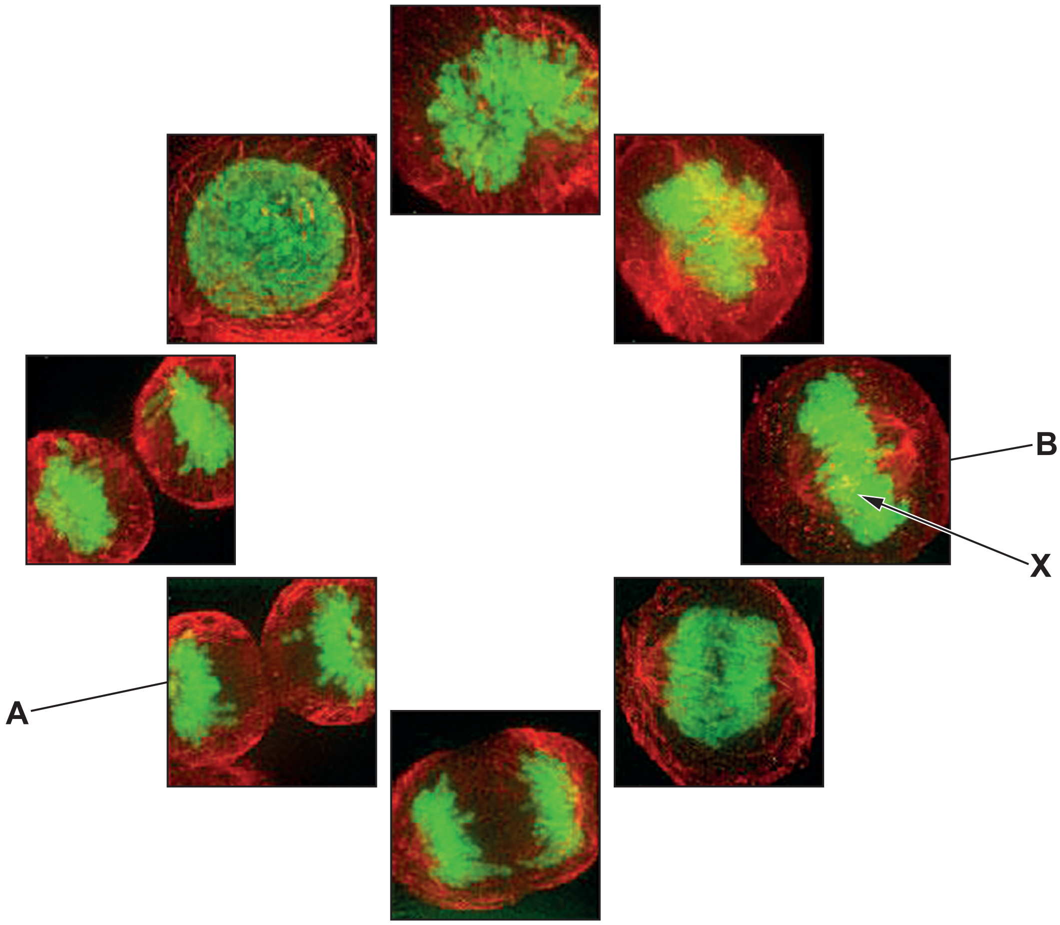
Answer **all** *questions.*

1. In 1951, Henrietta Lacks died of cervical cancer. A research scientist called George Gey, grew a sample of her tumour. He found that these cells multiplied rapidly and could be grown indefinitely in culture. They became the first immortal human cell line, which he named HeLa. HeLa cells are now grown in research laboratories in many different countries.

(a) The photomicrographs below show images of HeLa cells during different stages of the cell cycle. The cells have been stained with a dye, which causes the DNA to be visible.



(i) Name the stages shown in **A** and **B**. [2]

Stage **A**: ....

Stage **B**: ....

Examiner only

(ii) Name the structures, labelled **X**, that can be seen in the photomicrograph. [1]

....

(iii) Explain why some stain would be seen in other parts of the cell. [2]

....

....

....

....

(b) A biotechnology company that supplies HeLa cells to laboratories states that the number of HeLa cells double every 19 hours. They suggest that the starting culture of cells should have a density of 30 000 cells cm^{-3} and that the cells should be sub-cultured every 4 days.

Calculate the density of the cells that would be present after 4 days; give your answer in cells cm^{-3} in standard form. [3]

2400U101
03

Cell density = ... cm^{-3}

8